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Examining Program Quality Disparities Between Urban and Rural Kindergartens in China: Evidence From Zhejiang

Bi Ying Hu[a], Yisu Zhou[a], Kejian Li[b] & Sherron Killingsworth Roberts[c]
[a] University of Macau, Taipa, Macao
[b] Zhejiang Normal University, Hangzhou, China
[c] University of Central Florida, Orlando, Florida
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Bi Ying Hu and Yisu Zhou
University of Macau, Taipa, Macao

Kejian Li
Zhejiang Normal University, Hangzhou, China

Sherron Killingsworth Roberts
University of Central Florida, Orlando, Florida

In recent years, the Chinese government has initiated a national plan to universalize quality kindergartens for all age-eligible Chinese children. Kindergarten is the main form of early childhood education and care (ECEC) service across China. However, the government faces two thorny issues on the journey toward realizing educational equity for some of the nation's most vulnerable children. The first issue is the critical shortage of kindergarten programs. The second issue is a widening gap in quality between rural and urban kindergarten programs. Therefore, this study examines the quality of 91 kindergarten programs across different locations and funding sources in Zhejiang Province based on a stratified random selection method to gain insights into the aspects and magnitude of quality disparities in programs serving children from the richest and most powerful urban residents to the most disadvantaged village families. Program quality was rated utilizing eight subscales of the Chinese Early Childhood Programs Rating Scale, modified from Harms, Clifford, and Cryer's (2005) Early Childhood Environment Rating Scale. Finally, recommendations for future studies and policy formation are provided to enhance and perhaps hasten the process of universalizing quality kindergartens in the increasingly market-based context of modern China.

Keywords: Chinese kindergarten, program quality, education equity, preschool legislation, ECERS-R, CECERS

Alarming research findings have convinced policymakers worldwide of the significance of a universal provision of quality early childhood education and care (ECEC) in enhancing children's school readiness and achievement, particularly for children from disadvantaged backgrounds (Burger, 2010; Engle et al., 2007; Mwaura, Sylva, & Malmberg, 2008; Spodek & Saracho, 2006; Sylva et al., 2006). In developed countries, governments have willingly accepted findings from

Submitted October 15, 2012; accepted August 26, 2013.
Address correspondence to Bi Ying Hu, Faculty of Education, University of Macau, Av. Padre Tomas Pereira, Taipa 999078, Macao. E-mail: hubiying@umac.mo

empirical studies to guide policy formation to ensure access to a quality ECEC to all children through a range of social and educational services and support (Fenech, 2011). For instance, the U.S. government guarantees the right of access to ECEC to children from low-income families and those with disabilities through the provision of the federally funded Head Start and Early Head Start programs. In China, no current law ensures children will receive basic ECEC services, even with minimum quality (Hu & Li, 2012; Jiang, 2008). The most prevalent form of ECEC programs serving children age 3 to 6 are called kindergartens, or *you'eryuan* in Chinese. In recent years, the critical shortage of ECEC services has become a growing national crisis. Chinese ECEC scholars refer to this crisis as *ruyuannan*—meaning "you have to fight to get into a kindergarten." According to the Ministry of Education (MOE) of the People's Republic of China (2010), a total of 150,420 registered kindergartens currently serve only about half of the 100 million children of eligible age (i.e., ages 3 – 6). Out of these existing programs, 35,845 kindergartens serve urban families, 42,987 serve a growing number of families in counties and towns, and the remaining 71,588 kindergartens serve rural families. Despite the fact that rural kindergartens outnumber urban kindergartens by almost twofold, about 57% of rural children currently have no access to any form of ECEC (Rao, Sun, Zhou, & Zhang, 2012).

Evidently, a challenge remains for the government to institute a funding mechanism that would allow everyone much-needed opportunities for ECEC services. However, a greater challenge lies in developing ways to monitor and promote the quality of programs that serve the most disadvantaged rural children. Rao et al. (2012) conducted the first empirical study on Chinese rural children's educational attainment and their experiences with early education and discovered kindergarten programs, preprimary classes, sit-in experiences, and no preschool experiences. Findings suggest that Chinese rural children made higher mathematics and literacy achievement scores when they attended developmentally appropriate kindergarten programs, and children who had no preschool experiences reported lower achievement scores. Rao et al., therefore, recommended scaling up Chinese rural ECEC by expanding 2-year kindergarten programs and improving the teacher-child ratio and quality of these programs (i.e., quality learning environments and developmentally appropriate curriculum and teaching). Assuming that the number of ECEC programs will be significantly rising in the next decade so as to overcome the persistent challenge of urban-rural education quality disparity, local and national governments must begin to gather a comprehensive picture of urban and rural ECEC. Therefore, the authors in this study attempt to answer the following research questions:

1. Are there any differences in ECEC program quality in Zhejiang province based on locations of program (urban, county and town, and rural area) and funding sources (public and private) as measured by eight of the nine subscales of the Chinese Early Childhood Environment Rating Scale (CECERS), modified from Harms, Clifford, and Cryer's (2005) Early Childhood Environment Rating Scale (ECERS-R)?
2. Does the student-teacher ratio, percentage of teachers with a college degree, or the percentage of teachers with a professional license make a difference in the quality of rural ECEC programs?

Before examining the data in regard to these questions, the cultural and historical context of ECEC programs as reported in the literature will be explored below.

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THE URBAN-RURAL DISPARITY IN ECEC PROGRAM QUALITY

China has struggled with social consequences resulting from the ill-balanced socioeconomic development among Eastern and Western regions and between urban and rural areas (Law & Pan, 2009). Education inequity is one of those consequences, manifested in rural children's limited access to ECEC services with minimum quality provisions (Y. H. Li, 2000; Shi & Niu, 2007; Tang, 2005). Compared to urban kindergartens, many of which are sponsored by government-affiliated agencies, rural ECEC programs typically rely on private providers. The infrastructure, teacher quality, and instruction quality can vary substantially among these providers. Most recently, two empirical studies found rural Chinese children significantly lagging behind their urban counterparts in mathematics (Pan, Qiu, Zhou, & Liu, 2011) as well as in language and early literacy (Qin, Shi, & Liu, 2011) competencies even before they enter primary schools. Although the long-term effect of this urban-rural gap is yet to be understood systematically, some evidence suggests it can affect educational attainment and college matriculation for rural students (Wang et al., 2011).

The poor early academic performances of rural children compared to their urban counterparts can be attributed to a need for basic, minimum quality ECEC programs for rural children and a funding mechanism to execute and sustain these programs (Chinese National Center for Education Development Research & Chinese National Commission for UNESCO, 2008). Chinese rural kindergartens include kindergartens at the township and village levels. Traditionally, local governments or government-sponsored corporations have offered public ECEC services to parents who are government employees as a type of social benefit. Before economic reforms were initiated in 1980, rural township/village committees reserved funding for ECEC services to support the rural economy and prepare rural children for compulsory education (Tang, 2005). Town/village committees, however, gradually lost their administrative power as China's social and economic transformation deepened in the 1990s to embrace a market-based economy model; therefore, they could no longer support ECEC services for rural children. Consequently, private providers, mostly in the form of family businesses, took over ECEC services and competed to survive in China's transitional market economy without any form of public support (Jiang, 2008; Law & Pan, 2009). For a long time, rural children and children from poor families, including migrant children, attended these family-based, private care facilities, which offered minimum resources and staff with no formal ECEC training (Hu & Szente, 2009; Jiang, 2008). The available literature has generally reported unacceptable to poor quality of ECEC programs serving rural populations, especially those from economically disadvantaged or remote areas (Hu & Li, 2012; Hu & Roberts, 2013; Hu & Szente, 2009).

CONCERNS FOR THE GROWING PRIVATE KINDERGARTENS IN RURAL AREA

The public/private factor adds another layer of complexity to the urban-rural disparity in ECEC program quality. Public kindergartens make reference to ECEC programs that are fully funded and operated by government, including provincial and local departments of educational, governmental agencies, and/or public corporations. According to the Ministry of Education of the People's Republic of China (2010), these public providers operate 29.5% of the total ECEC

services, whereas private providers (i.e., for-profit businesses) operate 68.0% of them. Other organizations, often charitable organizations, operate the remaining 1.5% of ECEC services. ECEC is the most underfunded subsector, as it only received 1.3% of the total budget for education, which is extremely low by international comparison (usually 6%–8% of the total education budget). Furthermore, the majority of the total ECEC funding goes to fewer than 30% of the programs, which primarily serve children who are already advantaged as a result of their social class (Zhao & Hu, 2008). Researchers have consistently reported that public programs tend to produce higher quality services than their private counterparts (Cleveland, Forer, Hyatt, Japel, & Krashinsky, 2007; Hu & Szente, 2009; Pan, Liu, & Lau, 2010). The higher the quality rating of a kindergarten, the more likely that kindergarten is to (1) receive recognition and financial incentives from the MOE and community, (2) enroll children from families that have greater political power and higher socioeconomic status, and (3) recruit and retain well-trained, experienced, certified teachers (Hu & Li, 2012).

The fact that for-profit businesses provide 68% of the overall ECEC services to Chinese families, and provide an even greater percentage in rural areas especially to disadvantaged families, is of great concern to the Chinese government. For instance, 76.06% of rural ECEC services are private in Zhejiang Province. In recent years, China has primarily adopted a market-based model of ECEC, in which competition and choice are thought to ensure the model's well-being. However, the current ECEC institutions cannot meet society's demands, as many parents do not have equal access to quality ECEC services because of a shortage of providers. The critical shortage of service providers reduces the competitiveness of the business. In other words, many ECEC providers, particularly for-profit ones, are not motivated to improve the quality of services because most of them have a long waiting list of children applying for admission. Also, parents as the consumers often lack time to seek out and evaluate ECEC services opportunities to observe the quality of the child care they use (Cryer & Burchinal, 1997; Moss, 2009). Most rural parents tend to have a lower level of education and less income, and their lack of accurate information often leads to poor judgment concerning the quality of ECEC services. One may infer that rural parents are less likely to advocate for quality ECEC for their children due to a number of reasons, which prompts the government to make rural areas a priority when distributing public funding for ECEC programs, with the most economically disadvantaged villages receiving the most public funding and other resources, such as experienced teachers and lower student-teacher ratios.

NEW RAYS OF HOPE FOR RURAL CHILDREN

Now that the Chinese government has recognized the negative impact of "scarce rural children's quality education for the development of economy" on rural children (Chinese National Center for Education Development Research & Chinese National Commission for UNESCO, 2008), and to close the widening gap between rural and urban children, the government has made a commitment to universalizing quality ECEC for all children. In the *Compendium for China's Mid- and Long-Term Education Development*, a significant document that symbolizes China's national movement toward educational equity, the Central People's Government (2010) declared a 10-year plan for equalizing educational opportunities for all Chinese children. Specifically, the government set goals to ensure that all Chinese children, urban and rural, receive one year of ECEC by 2020. As a result of this ambitious plan, one can witness an increased number of rural

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ECEC programs receiving various forms of federal support (Ministry of Education of the People's Republic of China, 2009, 2010).

The Chinese government, however, continues to face the greater challenge of increasing the quality of programs serving rural children. The complexity of the Chinese socioeconomic system makes it difficult to adequately address the needs of different forms of ECEC programs, considering the program locations (e.g., urban vs. rural) and funding sources (e.g., public vs. private) as well as the populations they serve (e.g., residential vs. migrant). To adequately and effectively distribute resources to ensure a fair start for every child, local governments must have reliable tools to monitor and evaluate the quality of both urban and rural ECEC programs. Hence, this study explores and reports the current status quo of kindergartens in the Zhejiang Province in China, uncovering urban-rural disparity as indicated by eight dimensions of ECEC program quality. Although researchers, practitioners, and policymakers are aware of urban-rural disparity experientially, no studies to date have examined program quality as well as underlying, contributing factors. This study seeks to provide a detailed report concerning urban-rural ECEC program disparity so as to provide constructive feedback to Chinese policymakers working to improve rural ECEC services.

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METHOD

Participants

This study examined 91 kindergartens from six cities in Zhejiang, with two cities representing each of three levels of economic development: high, middle, and low. The province of Zhejiang currently provides ECEC services to 1,043,962 children age 3 to 6 through a total of 6,482 registered kindergartens. The 91 kindergartens from these six cities were stratified on the basis of location (urban, suburban, township, and village) and according to the funding source (public and private). A sample of 178 classrooms was randomly selected from within the 91 kindergartens for inclusion in this study for the following age groups: 45 classrooms for 3- to 4-year-olds, 53 classrooms for 4- to 5-year-olds, and 74 classrooms for 5- to 6-year-olds. Specifically, researchers selected between one or three classes (usually two) within each kindergarten to observe the program quality. The remaining six classrooms were classrooms for mixed ages. Table 1 shows the quality levels of the kindergartens as rated by the Chinese MOE. According to Table 1, the Ministry of Education of the People's Republic of China confirmed 13 out of 17 kindergartens (76.47%) rated as Level One or highest quality and were located in urban areas, whereas none of the kindergartens in rural towns or village areas were rated as Level One. Also, 15 out of the 17 (88.23%) Level One kindergartens were public; only one Level One kindergarten in an urban area was private.

Measure of Program Quality per Chinese Early Childhood Programs Rating Scale

The CECERS is a Chinese adaptation of Harms et al.'s (2005) Early Childhood Environment Rating Scale (ECERS-R) used for measuring the quality of ECEC program quality. The Chinese version of ECERS-R was first translated by Guo-Li and Cheng (2006) and published by Taiwan

TABLE 1
Quality Ratings of Participating Kindergartens Based on Location and Funding Source

	Locations				*Funding Source*	
Quality Rating	*Urban*	*County*	*Township*	*Village*	*Public*	*Private*
Level One	13	4	0	0	15	2
Level Two	14	6	12	1	19	13
Level Three	7	10	11	4	8	24
Permitted	0	3		7	2	8
Total	33	23	23	12	44	47

Note. The quality of kindergartens based on Zhejiang province's kindergarten quality rating system is in the following order from the highest rating to the lowest: Level-One, Level-Two, and Level-Three. Permitted means licensed kindergartens without ratings.

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Psychology Press in the same year. The current adaptation began by developing the trial version of the CECERS based on a review of existing literature on (1) ECEC quality evaluation in international contexts (e.g., Fenech, 2011; Lambert et al., 2008; Mathers, Linskey, Seddon, & Sylva, 2007; Sylva et al., 2006), (2) kindergarten quality ratings systems and standards in China (K. J. Li, 2010), and (3) published measures, such as Liu and Pan's (2008) Kindergarten Environment Rating Scale (KERS)—the only measure for quality created in Chinese contexts—and La Paro, Pianta, and Suthlman's (2004) Classroom Assessment Scoring System (CLASS). Also, in an earlier study, the authors explored the quality of 105 ECEC programs in Zhejiang using the Chinese version of the ECERS-R in Fall 2010. Specifically, using the ECERS-R, the authors explored how accurately the province of Zhejiang's Kindergarten Quality Rating System (KQRS) differentiated levels of quality. Findings uncovered a positive degree of congruence between underlying concepts of quality for the ECERS-R and Zhejiang's KQRS, which provided insights for revisions that enhanced its cultural authenticity, and resulted in the CECERS (see the Appendix for items measured in the CECERS).

With the permission of the ECERS-R authors, the CECERS kept the same 7-point Likert-type rating scale, with 1 representing *inadequate quality* and 7 representing *excellent quality*. Then, the researchers adapted the idea of using subitems from Liu and Pan's (2008) Kindergarten Environment Rating Scale (KERS). For instance, the first item Furnishing for Play and Learning has four subitems to assess size and structure, basic furnishings, safety, and cleanliness and maintenance. Most importantly, the researchers took into consideration the sociocultural, economic, and policy influences on ECEC in China, such as how Chinese teachers have traditionally utilized whole-group teaching as a cultural approach to curriculum planning and delivery and for the organization of learning environments. Thus, the subscale of Whole-group Instruction was added by the researchers to specifically measure the quality of group teaching in kindergartens.

The addition of Whole-group Instruction is necessary because, unlike Western cultures that value individualization and initiative, free choice, and democratic student-teacher relationships, Chinese culture values teacher-directed learning and showing respect for the teacher through obedience. Despite the Chinese government's efforts to integrate Western theories of developmentally appropriate practice (DAP) (Copple & Bredekamp, 2009) concepts into ECEC

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curriculum development and teaching (Hu, 2014), Chinese ECEC programs show strong contextual and cultural needs to seek a balance between teacher-directed and child-centered learning experiences, a balance between whole-group instruction and creative play, and a balance between individual and free choice activities. As a result, the quality indicators of CECERS reflect these three balances, as the major differences compared to the quality indicators outlined in the ECERS-R.

Many CECERS items share the same name with the ECERS-R, yet their scoring indicators, criteria, and examples may be quite different. Also, the trial version of the CECERS used in this study has nine subscales, with 63 items with more than 200 supporting subitems. The nine subscales are Space and Furnishings, Personal Care Routines, Curriculum Planning and Implementation, Group Teaching, Activities, Language and Reasoning, Guidance and Interactions, Parents and Staff Support, and Support for Inclusion Services. However, because the subscale of Support for Inclusion Services was not used in this study, given that most Chinese ECEC programs have not fully committed to serving children with special needs, the total subscales in the CECERS in this study remain at eight (Hu & Szente, 2010).

To further ensure the cultural authenticity of the CECERS, the researchers solicited feedback from 176 Chinese expert practitioners (model teachers who were recommended by the local [DOE] to mentor other teachers) and directors, to review the content validity of the CECERS for Chinese kindergartens. Overall, these reviewers highly supported the CECERS as a valuable and quality measure and provided salient revisions (Hu, Vong, Chen, & Li, 2014). Guided by the framework of broad conceptualization of validity and validation as advocated by Messick (1989), Li and his colleagues recently completed a validation study of the CECERS (Li, Hu, Pan, Qin, & Fan, 2014). In this validations study, evidence in a variety of forms was presented, including content validity considerations (e.g., measuring socially and culturally relevant domains), measurement reliability considerations (e.g., inter-rater reliability, internal consistency reliability, multivariate generalizability), and measurement validity considerations (criterion-related validity, concurrent validity, internal structure based on factor analysis) to provide empirical support for the use and interpretation of CECERS scores as quality indicators of preschool programs in the Chinese social and cultural contexts.

Data Collection

In summer 2011, a team of 28 research assistants (RAs), all master's level students majoring in ECEC at Chinese universities, were trained to serve as raters of the CECERS. The training consisted of 4 days for intensive study of the instrument and one week of field rating until all RAs reached an inter-rater reliability of at least .85 with the trainer. These raters implemented the trial version of CECERS during fall 2011, based upon the feedback from expert teachers and directors. During the spring of 2012, the 28 trained raters implemented a final version of CECERS in 178 classrooms. Two raters scored each classroom to ensure the reliability of the adapted ECERS-R scores. Specifically, the two RAs scored each classroom independently, then shared, compared, and consolidated their findings and finally submitted a single scored report as a team. The inter-rater reliability analysis using the weighted kappa statistics showed 13 items with outstanding agreement (kappa statistics over 0.8), 37 items with substantial agreement (kappa statistics between .6 and .8), and one item (i.e., classroom culture) with moderate agreement (Item 36) (kappa statistic at .505) (Landis & Koch, 1977).

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RESULTS

Comparison of Teacher Information

Table 2 shows basic teacher information for public and private kindergartens separately in relation to their locations. Student-teacher ratio and demographic information are important variables affecting the structural quality of each program.

According to Table 2, urban public kindergartens offered the lowest average observed student-teacher ratio (17:1), whereas village private kindergartens offered the highest (27:1). Urban public kindergarten teachers, on average, have five more years of teaching experience than village private kindergartens (from 12.99 – 7.55 years). Another important factor that may be considered is *bianzhi*. Awarded by the local DOE, *bianzhi* refers to whether a teacher has an officially established post, which is similar to a tenured position and also ties the teacher to important social benefits. Teachers without bianzhi not only earn less salary, but also receive a significantly lower amount of subsidies and other benefits (Hu & Roberts, 2013). Table 2 shows the reality that no rural village kindergarten teachers possess bianzhi; this remains true for public and private village programs. Another important designation is *zhicheng*, which is a professional rating awarded by the local DOE based on a teacher's degree and professional performance, a very important status that affects a teacher's salary and benefits. Moreover, almost 60% of public urban teachers have primary advanced zhicheng and only 6.3% of urban public teachers are without zhicheng. However, none of the private village teachers has earned any type of zhicheng.

The teachers' first degrees are reported because, in China, the majority of kindergarten teachers receive their highest degrees through nonpaid professional development for earning zhicheng or moving up the rank. In China, one's first degree indicates his or her educational capabilities more sensitively than the highest degrees in many regards. Table 2 also reveals that private rural kindergartens hired 99% of teachers whose first degree was at or below high school equivalency. On average, urban public teachers have started teaching 5 years earlier than private village teachers. Still, more than 34% of urban public teachers are teaching with an associate degree, significantly higher considering the fact that teachers' first degrees are being compared. On the other hand, about 25% of the public rural village kindergartens teachers now have entered the profession with an associate's degree, which is a promising increase.

Overall Performance on the CECERS

The overall average score for the 178 classrooms observed was 3.32 out of 7.0, indicating a quality of slightly above minimum on a scale that equates a score of 1 with *inadequate quality*, 3 with *minimum quality*, 5 with *good quality*, and 7 with *excellent quality*. However, Table 3 presents the mean score for each subscale across the 178 kindergarten classrooms in relation to their funding sources and locations.

Across all eight subscales explored for this study, public kindergartens were rated higher compared to private kindergartens and urban kindergartens were rated higher compared to village (rural) kindergartens. Not only do public and urban institutions score higher across all subscales, they were rated higher on the entire scoring distribution as shown by higher percentile statistics. The differences related to funding source were smaller than those related to locations, however.

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TABLE 2
Basic Teacher Information for Kindergartens of Different Location and Funding Source

		Urban		*County*		*Township*		*Village*	
		Public (n = 36)	*Private (n = 40)*	*Public (n = 24)*	*Private (n = 18)*	*Public (n = 29)*	*Private (n = 16)*	*Public (n = 4)*	*Private (n = 11)*
Bianzhi (%)		70.6	10.3	87.0	17.7	71.4	0		0
Zhigezheng (%)		56	48	53.2	34	51.5	36.7	57.9	15.4
Zhichen (%)	None	6.3	32.4	8.7	68.8	40.7	64.3	50	100
	Primary/second	3.1	8.8	4.3	6.3	11.1	21.4	25	0
	Primary/first	28.1	47.1	34.8	25	40.7	14.3	25	0
	Primary/advanced	59.4	11.8	52.2	0	7.4	0	0	0
	Middle/advanced	3.1	0	0	0	0	0		0
First degree (%)	Junior high	0	0	0	0	0	0	0	18.2
	High school	5.9	7.9	8.7	11.1	14.8	12.5	25.0	27.3
	Vocational	0	18.4	0	22.2	3.7	37.5	0	27.3
	Teacher's vocational	58.8	36.8	56.5	50.0	33.3	25.0	50.0	18.2
	Associate vocational	5.9	10.5	0	0	11.1	6.3	0	9.1
	Associate's	29.4	26.3	26.1	16.7	33.3	18.8	25.0	0
	Bachelor's	0	0	8.7	0	3.7	0	0	0
	Master's	0	0	0	0	0	0	0	0

	M	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*
Age	33.94	5.34	30.18	6.28	32.83	5.64	29.33	5.12	30.46	6.09	28.87	6.44	30.25	9.25	31.18	6.68
Years of teaching	12.99	6.71	9.13	6.79	11.90	7.11	7.61	5.02	7.30	4.62	7.86	5.78	9.00	10.44	7.55	5.81
Student teacher ratio	17.84	6.28	18.25	6.37	23.28	8.21	25.03	9.50	22.68	5.71	23.38	9.02	22.75	11.95	27.32	9.53

Note. Banzhi is a tenure status awarded by the local educational departments. Zhichen is a professional rating, based on observed performance, for which a teacher may apply to be evaluated. The rank of zhichen from the lowest to the highest: primary second, primary first, primary advanced, middle school advanced.

TABLE 3a
Description of Adapted Early Childhood Environment Rating Scale Rating Score by Funding Sources

Domain		*M*	*SD*	*25th percentile*	*Median*	*75th percentile*
Space and furnishings	Total	3.39	1.11	2.38	3.50	4.22
	Public	3.76	1.03	2.88	3.88	4.50
	Private	2.96	1.05	2.00	3.00	3.88
Personal care and routines	Total	3.57	1.15	2.67	3.67	4.33
	Public	3.89	1.12	3.00	3.83	4.67
	Private	3.20	1.07	2.33	3.00	4.00
Curriculum planning and implementation	Total	3.03	1.03	2.20	3.00	3.80
	Public	3.38	0.98	2.80	3.40	4.00
	Private	2.63	0.93	2.00	2.60	3.40
Whole-group instruction	Total	3.50	1.01	2.86	3.50	4.14
	Public	3.70	1.03	2.86	3.86	4.43
	Private	3.26	0.93	2.71	3.29	3.86
Activities	Total	2.47	0.93	1.63	2.44	3.25
	Public	2.81	0.89	2.00	3.00	3.50
	Private	2.07	0.80	1.38	2.00	2.88
Language-reasoning	Total	3.00	0.89	2.25	3.00	3.50
	Public	3.26	0.88	2.75	3.25	3.75
	Private	2.69	0.79	2.25	2.75	3.25
Guidance and interaction	Total	3.40	1.05	2.75	3.25	4.25
	Public	3.61	1.08	2.80	3.60	4.40
	Private	3.16	0.96	2.60	3.00	3.75
Parent and staff	Total	4.10	1.09	3.20	4.00	4.83
	Public	4.40	1.08	3.50	4.50	5.33
	Private	3.75	1.00	3.00	3.60	4.67

Comparing subscale mean scores based on locations. A regression model was used to compare subscale rating scores among schools at various locations. We first standardized rating scores across classrooms at different locations for each subscale. Then, we aggregated classroom rating scores to the school level. Next, we used a linear regression model with one categorical predicator, school location, to test the mean differences. The results are shown in Table 4.

Overall, urban kindergartens scored higher compared to kindergartens in other locations. Their advantage appears universal across eight subscales and is statistically significant. Table 4 also shows that the rating scores drop as locations move from urban to country, township, and rural village. Rural village kindergartens scored the lowest. If we were to compare rural village kindergartens to urban kindergartens, the rating score difference is close to one standard deviation on the Space and Furnishings, the Personal Care and Routines, and the Parent and Staff Support subscales. For the remaining five subscales, the differences were also greater than one half a standard deviation. Hence, the gaps are substantial. The rating scores between township and county kindergartens are close; the gap is about 0.2 to 0.46 standard deviations across the different subscales.

Comparing subscale mean scores based on kindergarten characteristics. Based on the first model, two school characteristics, private/public indicator and per-classroom input reported by the kindergarten director, were added. The per-classroom input is the amount of

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TABLE 3b
Description of Adapted Early Childhood Environment Rating Scale Rating Score by Urban and Rural

Domain		*M*	*SD*	*25th Percentile*	*Median*	*75th percentile*
Space and furnishings	Total	3.39	1.11	2.38	3.50	4.22
	Urban	4.00	1.03	3.50	4.11	4.67
	Village	2.16	0.59	1.88	2.00	2.38
Personal care and routines	Total	3.57	1.15	2.67	3.67	4.33
	Urban	4.29	1.06	3.33	4.33	5.17
	Village	2.27	0.74	2.00	2.33	2.50
Curriculum planning and implementation	Total	3.03	1.03	2.20	3.00	3.80
	Urban	3.47	0.98	2.80	3.40	4.00
	Village	2.26	0.81	1.60	2.20	3.00
Whole-group instruction	Total	3.50	1.01	2.86	3.50	4.14
	Urban	4.06	0.94	3.57	4.00	4.71
	Village	2.79	0.82	2.29	2.86	3.14
Activities	Total	2.47	0.93	1.63	2.44	3.25
	Urban	2.92	0.83	2.13	3.00	3.50
	Village	1.61	0.70	1.13	1.25	1.88
Language-reasoning	Total	3.00	0.89	2.25	3.00	3.50
	Urban	3.45	0.87	3.00	3.50	4.00
	Village	2.24	0.66	1.75	2.25	2.75
Guidance and interaction	Total	3.40	1.05	2.75	3.25	4.25
	Urban	3.97	1.09	3.20	4.20	4.80
	Village	2.64	0.49	2.60	2.75	2.80
Parent and staff	Total	4.10	1.09	3.20	4.00	4.83
	Urban	4.79	0.94	4.00	4.83	5.33
	Village	2.84	0.69	2.60	2.80	3.20

monetary resources the school is allocated to each classroom, keeping in mind that this variable is constant within each school. The results are shown in Table 5.

The earlier-reported urban-rural village differences still existed but were reduced. The variation is partially explained by the two variables we added. Private kindergartens on average scored 0.2 to 0.6 standard deviations below public kindergartens across the subscales (except for the Whole-group Instruction subscale) after controlling for monetary input. Private village kindergartens scored lower on the rating scale. This pattern holds for all subscale ratings. Although the small sample size limited our ability to statistically test the differences across particular combinations of school location and funding sources, our exploratory analysis suggests that the differences between public rural village kindergartens and private urban kindergartens have a much smaller gap compared to the differences between public and private rural village kindergartens. For instance, public rural village kindergartens are 0.17 standard deviations below public urban kindergartens in the Space and Furnishings subscale in mean comparison, wherein the difference is 1.16 between public and private village kindergartens. This pattern holds for all the subscales. Also, this pattern suggests within-region between-school disparity in quality ratings is equally, if not more, important as between-region differences. Per-classroom financial input is positively associated with rating scores, although the coefficients are smaller. One unit increase

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TABLE 4
Compare Kindergartens of Different Locations

	Subscale							
Variable	*Space and furnishings*	*Personal care and routines*	*Curriculum planning and implementation*	*Whole-group instruction*	*Activities*	*Language-reasoning*	*Guidance and interaction*	*Parent and staff support*
Location[a]								
County	−0.81***	−0.86***	−0.74**	−0.80**	−0.77**	−0.73**	−0.82***	−0.82***
	(0.24)	(0.22)	(0.25)	(0.25)	(0.25)	(0.24)	(0.24)	(0.22)
Township	−0.93***	−1.11***	−0.80**	−0.90***	−0.75**	−0.98***	−0.99***	−1.21***
	(0.24)	(0.22)	(0.25)	(0.24)	(0.25)	(0.24)	(0.24)	(0.22)
Village	−1.46***	−1.68***	−1.18***	−1.21***	−1.25***	−1.28***	−1.26***	−1.75***
	(0.29)	(0.27)	(0.31)	(0.30)	(0.31)	(0.30)	(0.30)	(0.27)
Constant	0.63***	0.72***	0.55***	0.59***	0.55***	0.60***	0.62***	0.74***
	(0.15)	(0.14)	(0.16)	(0.16)	(0.16)	(0.15)	(0.15)	(0.14)
N	91	91	91	90	91	91	91	91

Note. Standard errors in parentheses.
[a]Urban schools as reference group.
$^{**}p < .01$. $^{***}p < .001$.

(10,000 renminbi per year) on the input variable is associated with a 0.5 to 0.11 increase in rating scores after controlling for location and funding sources.

Comparing subscale mean scores based on teacher characteristics. Table 6 provides three indicators of school-level teacher characteristics: student-teacher ratio, percentage of teachers in the school with licensure, and percentage of teachers with bianzhi, or an officially established post. The urban-rural gap remains significant for four subscales after controlling for teacher characteristics. Student-teacher ratio had a negative and statistically significant coefficient, showing that larger classrooms typically rated as lower quality. If the student-teacher ratio increases by 10, the school will suffer 0.4 to 0.6 standard deviations loss on the CECERS quality rating, noting a substantial difference. The percentage of teachers with a teaching license is positively correlated with rating scores. These scores suggest that if schools were to replace 10% of their unlicensed teachers with licensed teachers, the quality ratings would increase by 0.2 standard deviations across all subscales. The proportion of teachers with a government-established post did not correlate with the rating scores. Because some of the teacher variables suffered from missing data, these findings may only be interpreted as exploratory, of course.

DISCUSSION

This study examined the long-standing disparity of kindergarten program quality between urban and rural locations as measured through structural variables (e.g., student-teacher ratio and teacher qualification) and the CECERS using eight subscales or dimensions, omitting the ninth subscale related to offering inclusive services to children with special needs. The findings, overall, support Pan et al.'s findings (2010), which reported that the quality of Chinese kindergartens

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TABLE 5
Comparing Kindergartens Controlling for School Characteristics

	Subscale							
Variable	*Space and furnishings*	*Personal care and routines*	*Curriculum planning and implementation*	*Whole-group instruction*	*Activities*	*Language-reasoning*	*Guidance and interaction*	*Parent and staff support*
Location[a]								
County	−0.53*	−0.62**	−0.60*	−0.55*	−0.59*	−0.49*	−0.53*	−0.57**
	(0.21)	(0.22)	(0.26)	(0.25)	(0.25)	(0.24)	(0.26)	(0.21)
Township	−0.61**	−0.90***	−0.71**	−0.64*	−0.57*	−0.77**	−0.71*	−0.89***
	(0.22)	(0.23)	(0.27)	(0.26)	(0.26)	(0.25)	(0.27)	(0.22)
Village	−0.99***	−1.26***	−0.98**	−0.93**	−0.95**	−0.95**	−0.95**	−1.45***
	(0.26)	(0.28)	(0.32)	(0.32)	(0.31)	(0.30)	(0.32)	(0.26)
Private school	−0.62***	−0.57**	−0.74***	−0.29	−0.74***	−0.68***	−0.41*	−0.60***
	(0.16)	(0.17)	(0.19)	(0.19)	(0.18)	(0.18)	(0.20)	(0.16)
Per-classroom input[b]	0.11***	0.07*	0.06+	0.05+	0.08**	0.08**	0.07*	0.08**
	(0.02)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.02)
Constant	0.26	0.57*	0.60*	0.38	0.46+	0.43+	0.34	0.50*
	(0.23)	(0.24)	(0.28)	(0.27)	(0.27)	(0.26)	(0.28)	(0.23)
N	79	79	79	79	79	79	79	79
Private urban vs. public rural	0.17	0.17	−0.25	0.37	−0.3	0.02	0.35	0.68
Public vs. private rural	1.16	1.36	1.08	0.76	1.2	1.01	0.77	1.16
Public rural vs. private township	−0.53	−0.56	−0.63	−0.4	−0.58	−0.56	−0.6	−0.66
Public rural vs. private county	−0.56	−0.39	−0.51	−0.53	−0.57	−0.5	−0.4	−0.49

Note. Standard errors in parentheses.
[a]Urban schools as reference group. [b]Unit in 10,000 reminbi.
$+p < .10$. $^{*}p < .05$. $^{**}p < .01$. $^{***}p < .001$.

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TABLE 6
Comparing Kindergartens Controlling for School-Level Teacher Characteristics

	Subscale							
Variable	*Space and furnishings*	*Personal care and routines*	*Curriculum planning and implementation*	*Whole-group instruction*	*Activities*	*Language-reasoning*	*Guidance and interaction*	*Parent and staff support*
Location[a]								
County	−0.29	−0.28	−0.20	−0.15	−0.26	−0.12	−0.14	−0.29
	(0.25)	(0.23)	(0.32)	(0.29)	(0.29)	(0.29)	(0.32)	(0.24)
Township	−0.42	−0.55*	−0.25	−0.46	−0.26	−0.41	−0.26	−0.66*
	(0.28)	(0.25)	(0.35)	(0.32)	(0.32)	(0.32)	(0.36)	(0.27)
Village	−0.79*	−0.87**	−0.36	−0.80*	−0.44	−0.61	−0.65	−1.33***
	(0.34)	(0.31)	(0.42)	(0.39)	(0.39)	(0.39)	(0.43)	(0.32)
Student-teacher ratio	−0.06***	−0.05***	−0.05**	−0.04**	−0.05**	−0.05**	−0.06**	−0.05***
	(0.01)	(0.01)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.01)
Percentage of teacher with an official license	0.02**	0.02***	0.02*	0.01*	0.02***	0.02***	0.02*	0.02***
	(0.01)	(0.00)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Percentage of teacher with bianzhi	−0.00	−0.01	−0.00	−0.01	−0.01*	−0.01	−0.01	−0.01
	(0.00)	(0.00)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.00)
Constant	0.79*	0.63+	0.65	0.73+	0.47	0.44	0.72	0.63+
	(0.36)	(0.32)	(0.45)	(0.41)	(0.42)	(0.41)	(0.46)	(0.34)
N	57	57	57	57	57	57	57	57

Note. Standard errors in parentheses.
[a]Urban schools as reference group.
$+p < .10$. $^{*}p < .05$. $^{**}p < .01$. $^{***}p < .001$.

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varies based on the location and funding source of each program, with an average rating of slightly above minimum level of educational quality. Specifically, the findings indicate a wide disparity in program quality between urban and rural kindergartens, wherein urban public programs received the highest quality rating in each dimension and rural village private kindergartens received the lowest quality rating. Public rural village kindergartens, due to recent investments from the Chinese governmental early childhood initiatives, are in a better position to catch up with the quality of public county and township kindergartens.

The most alarming finding is that, in fact, the quality of private rural village kindergartens was rated between inadequate and minimum despite those recent increases in budgets. As previously reported, private rural village kindergartens are serving children from the most disadvantaged rural families including those who are migrant families, as these programs tend to charge lower fees than public village kindergartens (Hu & Roberts, 2013; Hu & Szente, 2009). It is important to note that a large number of private rural village kindergartens are out of compliance with the local MOE for licensure (Hu & Szente, 2009), which makes them unregistered programs. In Zhejiang, internal records of 2011 show approximately 869 unregistered kindergartens serving 56,119 children. In many areas, families have no choice financially but to send their child to the cheapest unregistered kindergartens. One may appropriately speculate that the quality of unregistered private rural village programs could be even lower than those included in the study since they did not even meet minimum safety and health standards to be awarded licensures and thus be included in the study. Without a doubt, the quality of private rural village kindergartens is worrisome to ECEC policymakers and scholars. The key to closing the gap between rural-urban inequities in Chinese ECEC lies in increasing governmental investments in the lowest rated rural village kindergartens, many of which are private. The Chinese government might consider transforming all private rural village kindergartens into public facilities or semipublic programs and develop innovative policies to attract quality teachers and lower student-teacher ratios to scale up rural ECEC. At a minimum, rural village private kindergartens deserve equal public funding and staffing with quality teachers as their urban counterparts enjoys.

Our findings strongly support the notion of prioritizing public funding for rural village kindergartens, particularly targeting private institutions. First, results reveal a large gap between urban and rural, public and private kindergartens in the structural variables that previous studies have found most crucial to ensure program quality (i.e., teacher's education level and child-staff ratio) (e.g., Arnett, 1989; Howes, Phillips, & Whitebook, 1992). Table 2 showed that urban public kindergartens offer the best average observed student-teacher ratio (17:1), hire the most teachers with licensure (zhicheng and bianzhi), and attract teachers with the best educational backgrounds who tend to commit to the job lifelong. Conversely, private rural village kindergartens offer the most severe (27:1) student-teacher ratio, no bianzhi (which is determined by the local MOE), hire teachers with no license or zhicheng, and attract teachers with the lowest quality of training with a much higher turnover rate.

In summary, high-quality teachers are least likely to work in private rural village kindergartens, which serve the most disadvantaged children and which are provided little or no public funding or resources. Another piece of strong evidence that supports the notion of prioritizing funding and teacher support for private village kindergartens is that very few private kindergartens are motivated to make investments necessary to earn a higher quality rating (K. J. Li, 2010). Building a high-quality kindergarten, such as the Level-One programs rated by the DOE, is not seen to be cost effective for private service providers (Hu & Li, 2012; Tang, 2005). Recent results

from the 2009 round of Programme for International Student Assessment (PISA), which compared 15-year-olds internationally across mathematics, science, and reading, provides additional evidence of the effect of ECEC on high academic achievement. According to a PISA presentation (Schleicher, 2010), Shanghai had the highest scores among participating countries; however, about 10% of the variance was attributed to 15-year-olds who had one year of kindergarten over those who did not. China's intentions to universalize and offer high-quality kindergarten programs to all its citizenry has the potential to boost overall student achievement to even greater heights. The following section provides discussion regarding the differences of process quality that exists between urban and rural ECEC as measured by the CECERS that could play an important role in achieving high-quality ECEC for all children in China.

Significant Program Quality Disparity Between Urban and Rural Kindergartens

Table 3 provides the mean scores as calculated on the raw 7-point scale. The following four subscales offer insights into significant program quality disparity between urban and rural village kindergartens: Personal Care and Routines, Support for Parents and Staff, Space and Furnishings, and Guidance and Interactions. First, these results reveal that Personal Care and Routines is the dimension wherein the largest gap between urban ($M = 4.29$) and rural village ($M = 2.27$) kindergartens exist. The subscale Personal Care Routines measures how kindergartens provide routine care (e.g., mealtime, toilet time, naptime) in a culturally sensitive and developmentally appropriate way, using child-friendly, medically approved facilities, and caring procedures to ensure their safety and promote their health. Similar to the findings of Hu and Roberts (2013), the overall rural kindergartens' toilet equipment and their efforts for maintaining sanitary conditions are far from satisfactory in terms of minimizing the spread of germs in classrooms. The personal care situations were rated even lower among private rural village kindergartens. For instance, many rural village kindergarten staffs were observed to omit hand washing before and after meals, after playing with dirty materials outside, and even after using the toilet. Based on field observations, private rural village kindergartens are more likely to ignore children's physical and hygienic safety by (1) utilizing communal, ill-maintained squat toilets; (2) placing large motor equipment, such as climbing structures, on concrete floors without any padding; and (3) utilizing potentially hazardous equipment, such as hazardous trash present, outdoors and indoors. Furthermore, the teacher-student ratios (1:27) in private rural village kindergartens may help to explain the poor score in Personal Care and Routines. One important message for the local governments is to strengthen management in this area of quality care through mandatory training and inspections.

Support for Parents and Staff is another dimension where large gaps between urban ($M = 4.79$) and rural village ($M = 2.84$) kindergartens exist. This subscale measures how teachers collaborate with parents and how kindergartens conduct teacher evaluation and foster professional growth. Unlike urban well-resourced kindergartens that regularly conduct systematic self-evaluations, peer evaluations, and expert reviews of each teacher, rural village kindergartens lack the financial resources for these activities (Tang, 2005). Future research should seek to identify effective ways for rural teachers to effectively collaborate with each other and with rural parents who, unlike urban parents, tend to hold a lower level of education and socioeconomic status and are less aware of ways to enrich children's development and learning (Hu & Roberts, 2013; Yu & Liang, 2008). Therefore, one area for future research to address is the role of local educational departments in facilitating an innovative model of home/school partnerships for rural

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village kindergartens. The development of an effective model of home/school partnerships in rural locations is critical for motivating rural teachers and promoting the well-being of rural children and their families.

The subscale Space and Furnishings is another area of significance in the urban-rural ECEC program quality disparity. Although rural village kindergartens ($M = 2.16$) have more space for indoor and outdoor play than most urban kindergartens ($M = 4.00$), they were rated well below their urban counterparts. In this study, private village programs were rated significantly lower by observers, who could find no evidence of clearly defined learning centers (e.g., book area or block area) and outdoor play areas (e.g., areas for group exercises, large stationary equipment, and swings) in rural village kindergartens. Rural village programs rarely had the resources for creating learning centers. In general, rural kindergartens were equipped with far less quality furnishings for indoor and outdoor play, or with learning and routine care equipment that was in good repair. Most rural private kindergartens remain inadequately staffed with high turnover; thus, teachers were observed doing a poor job of maintaining the order and cleanliness of classrooms; this situation was rated lowest among private rural village kindergartens. Conversely, urban teachers, including most suburban (county) public teachers, were rated higher in setting up learning centers, utilizing diverse furnishings of good quality, and using furniture to enrich children's play, such as easels for painting. They were also rated higher because they were observed displaying children's work, connecting the learning environment to the monthly theme, and making accommodations for individuals in terms of Space and Furnishings.

Last, the subscale Guidance and Interactions also detected differences between urban ($M = 3.97$) and rural kindergartens ($M = 2.64$). In this study, urban kindergartens were more likely to be observed providing developmentally appropriate supervision and interactions. Rural village teachers, for public and private programs, were less likely to be rated higher for using praise to reward positive student behaviors and interactions. Observations revealed that private rural village kindergarten children (1) engaged more frequently in rough play and verbal conflict during play, (2) used polite words less frequently, and (3) had fewer opportunities to engage in collaborative play guided by the teacher. Our findings supported prior research (Hu & Roberts, 2013) in that the contextual constraints often fed by financial constraints and teacher training, such as lack of play materials and play areas, contributed to decreased positive guidance and interactions among rural village children attending private programs.

However, limitations to this study are important to consider. Because this study was limited to one province, it may not be generalizable to the whole of China. Another limitation of this study is the omission of unregistered private rural kindergartens within the sample, which indeed, upon examination, provide representation of the ECEC programs with the lowest quality who are serving the most disadvantaged children in China. Recognizing the fact that this study did not examine the programs of the lowest quality might urge Chinese ECEC policymakers and scholars to make efforts in future studies to develop effective improvement plans and supporting policies to address improving the quality of unregistered private village kindergartens.

CONCLUSION AND IMPLICATIONS FOR EDUCATIONAL POLICY

Closing the quality disparity between urban and rural ECEC services has become a pressing issue for Chinese governments who are seeking empirical evidence for future policy formation (Central

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People's Government of the People's Republic of China, 2010). Our findings, though only generalizable to the province of Zhejiang, provide insights related to the glaring disparities that exist alongside Zhejiang MOE's priorities to minimize the differences between public and private programs in both the structural and process aspects of program quality. First, the Zhejiang MOE must develop policies for not only recruiting, but also retaining high-quality teachers for all private kindergartens, particularly private rural village kindergartens. For example, the government must ensure that highly qualified teachers be deployed to rural teaching posts, thus increasing the percentage of rural private kindergarten teachers with a teaching license. The local education departments must utilize an effective system for evaluating teacher performances for private programs in a form that can motivate and inform teachers simultaneously while positively promoting their teaching competencies.

Next, the Zhejiang MOE should institute a funding mechanism to ensure that private kindergartens, particularly those in rural villages, possess equitable provisions in terms of furnishings, equipment, and educational materials for indoor and outdoor play, learning centers, and routine personal care. A reasonable student-teacher ratio for all types and locations of programs must be consistently mandated by the Zhejiang MOE, which also links positively to Zhejiang's rating system, KQRS. The Zhejiang's MOE is in a position to impose standards for structural features, such as staff-child ratio and teacher qualifications while providing public funding and support for teacher training. For instance, the MOE needs to have a monitoring or evaluation system to make certain that rural village private kindergartens are adequately staffed with trained personnel throughout the year, not just during the days of program evaluation.

No doubt, structural quality improvements are necessary and fundamental for building process quality and thus program quality. The key to the improvement of process quality and its effectiveness for children and their families is hiring high-quality teachers who understand the importance of planning effective learning environments, rotating educational and creative materials, and flexibly adapting curricula to facilitate developmentally appropriate activities based on the contextual needs of kindergarteners (LoCasale-Crouch et al., 2007). Our findings show that rural kindergarten teachers, particularly private village teachers, need systematic training on developmentally appropriate curricula and instruction for the cultural contexts they serve. This is also true for all levels of private kindergartens lagging behind public counterparts. Perhaps the widening socioeconomic gap between urban and rural village development and culture in modern China necessitates further research and training to adequately address the need for contextually enhanced teaching resources and authentic strategies to reflect and promote equal access to high-quality ECEC.

In conclusion, China's latest national initiative for pursuing educational equity in ECEC truly represents an opportunity for Chinese citizens to practice a form of democratic participation in making choices for the collective good of the state by affording every eligible child a quality ECEC, regardless of their birthplace or socioeconomic status. As Chinese leaders embark on this journey toward equitable education, this forward-thinking decision is worthy of high praise as it will elevate all of China's children in terms of their capabilities and their contribution to China's future growth and stability. Future research using a variety of methodologies needs to address how to (1) institute a funding mechanism that will give every rural child equal access to a high-quality public kindergarten, (2) recruit and retain high-quality teachers for rural village kindergartens, and (3) motivate private kindergartens serving the most economically disadvantaged population to improve the structural and process quality of their services. Our findings in

regard to public village kindergartens offer assurance that once these important schools are provided equal resources and structural provisions, these rural village kindergartens can start to make improvements in most aspects related to program quality.

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APPENDIX: DESCRIPTION OF ITEMS

Domain	*Item*
1. Space and furnishings	1.1 Indoor space
	1.2 Furniture for play and teaching
	1.3 Toilet
	1.4 Space and furnishing for nap
	1.5 Room arrangement for play
	1.6 Space and furnishing for comfort and privacy
	1.7 Decoration and child-related display
	1.8 Space and equipment for gross motor
	1.9 Provisions for staff
2. Personal care and routines	2.10 Greeting/departing
	2.11 Toileting/washing/drinking
	2.12 Meals/snacks
	2.13 Nap/rest
	2.14 Health practice
	2.15 Safety practice
3. Curriculum planning and implementation	3.16 Weekly curriculum plan
	3.17 Daily schedule
	3.18 Free play
	3.19 Group activities
	3.20 Outdoor activities
4. Whole-group instruction	4.21 Objectivities and content of teaching
	4.22 Emotional support
	4.23 Instructional design and organization
	4.24 Teaching process
	4.25 Instructional support
	4.26 Performance of children
	4.27 Classroom culture
5. Activities	5.28 Social role play/drama
	5.29 Blocks
	5.30 Find motor
	5.31 Math
	5.32 Nature/science
	5.33 Music/movement
	5.34 Art
	5.35 Sand/Water
	5.36 Use of multimedia and computer
6. Language-reasoning	6.37 Books and reading
	6.38 Encouraging children to communicate
	6.39 Informal use of language
	6.40 Concept developing and reasoning

(*Continued*)

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APPENDIX
(*Continued*)

Domain	*Item*
7. Guidance and interaction	7.41 Guidance and supervision
	7.42 Discipline
	7.43 Interactions between staff and child
	7.44 Interactions among children
	7.45 Respect and appreciation for differences
8. Parent and staff	8.46 Services for parents
	8.47 Home-school communication and collaboration
	8.48 Meeting appropriate needs of teachers
	8.49 Interactions and collaborations among teachers
	8.50 Monitoring and evaluation of teacher performance
	8.51 Support for teachers' professional development

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